

Lab Session 1: Python Basics

1. Write a program that calculates the area of a circle. The circle radius is entered by users.

Expected screen output:

```
Enter circle radius? 10
Circle area = 314.0
```

2. Write a program that converts Celsius ($^{\circ}\text{C}$) into Fahrenheit ($^{\circ}\text{F}$).

Expected screen output:

```
Enter the temperature in Celsius? 10
10 (C) = 50.0 (F)
```

3. Write a program that checks whether a number is prime or not.

Expected screen output:

```
Enter a number? 3
3 is a prime number
```

```
Enter a number? 6
6 is a NOT prime number
```

4. Write a program that checks whether a number is perfect or not.

Expected screen output:

```
Enter a number? 6
6 is a perfect number
```

```
Enter a number? 28
28 is a perfect number
```

```
Enter a number? 30
30 is a NOT perfect number
```

5. Write a program that asks user for their favorite color. Look for this color in the given list. If the color is found, print out the index of this color in the list. If not, print out “Sorry, I could not find your color”.

Expected screen output:

What is your favorite color? Red Your colod is at index 3 in my list
What is your favorite color? Green Sorry, I could not find your color

6. Using range(), create and print out the following sequence:

Name	Sequence
range1	0, 1, 2, 3, 4, 5, 6
range2	1, 4, 7, 10
range3	5, 4, 3, 2, 1
range4	6, 4, 2, 0, -2

7. Write a function that removes the dollar sign (“\$”) in a string, named *remove_dollar_sign*, takes 1 parameter: s, where s is the input string, returns the new string with no dollar sign in it.
8. Write a function that extracts the even items in a given integer list, named *extract_even*, takes 1 parameter: l, where l is the given integer list ([1, 4, 5, -1, 10] for example), returns a new list contains only even numbers ([4, 10] if the given list is [1,4,5,-1,10]).
9. Write a Python function to calculate the factorial of a number (non-negative integer). The function accept the number as an argument.
10. Write a Python function get out all of divisors of a number.
11. Write a Python program to compute the distance between two points.
12. Write a Python function to print out the following pattern with size m x n:

```
* * * * *
*           *
*           *
* * * * *
```